

[CO-1]

Assessment Tool ~ I

Name — N. Rishitha

Regno — 192425011

Course code — MLA0401

Course — Deep Learning

46
—
50

92%
marks

Assessment Tool-I

- ① A coin is tossed 10 times and the outcomes are
H H T H H H T H H T

Definition

Maximum likelihood estimation (MLE) is a statistical method used to estimate the value of an unknown parameter from observed data.

Formula $\hat{p} = \frac{\text{Number of successes}}{\text{Total Number of Trials}}$

Given Total Tosses = 10
Number of Heads = 7

$$\Rightarrow \hat{p} = \frac{7}{10} \Rightarrow \hat{p} = 0.7$$

$$\underline{p = 0.7 \text{ (70\%)}}$$

- ② out of 50 students, 42 passed the exam

Definition

MLE estimates the probability of success by dividing the number of successful outcomes

$$\hat{p} = \frac{\text{Number of successes}}{\text{Total observations}}$$

$$\hat{p} = \frac{42}{50}$$

$$\hat{p} = 0.84$$

The estimated probability that a student passes the exam is 84%.

- ③ A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective

Definition

ML estimates the probability of an event using the sample data

$$\hat{p} = \frac{\text{Number of Defective bulbs}}{\text{Total bulbs}}$$

Calculation $\hat{p} = \frac{8}{100} \Rightarrow \hat{p} = 0.08$

The estimated probability that a bulb is 0.08 (8%)

- ④ A biased die is rolled 60 times. The number 6 appears 18 times

$$\hat{p} = \frac{\text{Number of successes}}{\text{Total Trials}}$$

$$\hat{p} = \frac{18}{60} \Rightarrow \hat{p} = 0.3$$

The estimated probability of getting 6 is 0.3 (30%)

- ⑤ Given

Input (x) = 4

weight (w) = 2

Actual output = 10

Learning rate = 0.01

Gradient = 4

prediction formula: $y = wx$

①

$$y = 2 \times 4$$

$$y = 8 \Rightarrow \text{predicted output} = 8$$

(ii)

Calculate Error

$$\text{Error} = \text{Actual output} - \text{predicted output}$$

$$= 10 - 8 = 2$$

$$\text{Error} = 2$$

(iii)

update weight

$$\text{Net weight} = w - (\eta \times \text{Gradient})$$

$$= 2 - (0.01 \times 4) = 2 - 0.04 = 1.96$$

predicted output = 8, error = 2, updated weight = 1.96